C

SAAT / GAT QUANTITATIVE (ENGLISH) • UNIT 4.01

GAT Data Interpretation, Averages & Counting

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

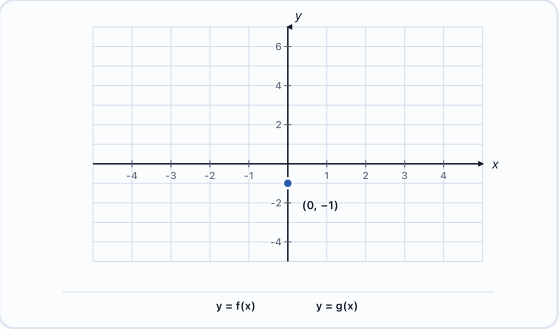
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 D	Q2 A	Q3 B	Q4 C	Q5 D
Q6 A	Q7 B	Q8 C	Q9 B	Q10 B
Q11 D	Q12 C	Q13 B	Q14 B	Q15 B
Q16 B	Q17 A	Q18 A	Q19 D	Q20 D
Q21 C	Q22 D	Q23 C	Q24 A	Q25 D

GAT Data Interpretation, Averages & Counting Practice Exercises

1 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 12x + 50$ in the domain of GAT Data Interpretation, Averages & Counting. By completing the square $f_1(x) = (x-h)^2 + k$, find the sum $k + f_1(8)$.



A 38

B 18

C 30

D 32 ✓

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (GAT Data Interpretation, Averages & Counting):

1. Complete the square: $x^2 - 12x + 50 = (x-6)^2 - 36 + 50 = (x-6)^2 + 14$, giving minimum value $k = 14$. **[M1]**
2. Evaluate at $x = 8$: $f_1(8) = (2)^2 + 14 = 18$. **[M1]**
3. Sum the two quantities: $k + f_1(8) = 14 + 18 = 32$. **[A1]**

- **Final Verified Answer:** 32.

2 MCQ

A mathematical model in GAT Data Interpretation, Averages & Counting is defined by the cubic function $y_2(x) = 2x^3 - 6x^2 + 15x - 11$. Calculate the exact gradient of the tangent to the curve at the point where $x = 3$.

- **A** 33 ✓
- **B** 69
- **C** 18
- **D** 21

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (GAT Data Interpretation, Averages & Counting):

1. Differentiate term by term using the power rule: $\frac{dy}{dx} = 6x^2 - 12x + 15$. **[M1]**
2. Substitute $x = 3$: $\left.\frac{dy}{dx}\right|_{x=3} = 6(9) - 12(3) + 15 = 33$. **[M1][A1]**

- **Final Verified Answer:** 33.

3 MCQ

In GAT Data Interpretation, Averages & Counting, two displacement vectors are given by $\mathbf{u}_3 = \langle 6, -8 \rangle$ and $\mathbf{v}_3 = \langle 9, 7 \rangle$. Compute the exact scalar value of $\mathbf{u}_3 \cdot \mathbf{v}_3 + \|\mathbf{u}_3\|^2$.

- **A** -2
- **B** 98 ✓
- **C** 100
- **D** 210

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (GAT Data Interpretation, Averages & Counting):

1. Compute the inner product: $\mathbf{u}_3 \cdot \mathbf{v}_3 = (6)(9) + (-8)(7) = 54 + (-56) = -2$. **[M1]**
2. Compute the squared norm: $\|\mathbf{u}_3\|^2 = 6^2 + (-8)^2 = 100$. **[M1]**
3. Combine: $-2 + 100 = 98$. **[A1]**

- **Final Verified Answer:** 98.

4 MCQ

The linear growth profile shown in the diagram for GAT Data Interpretation, Averages & Counting follows an arithmetic progression with first term $a_1 = 7$ and common difference $d = 6$. Find the exact sum S_{15} of the first 15 terms.

- **A** 91
- **B** 825
- **C** 735 ✓
- **D** 1365

WORKED SOLUTION & VERIFICATION

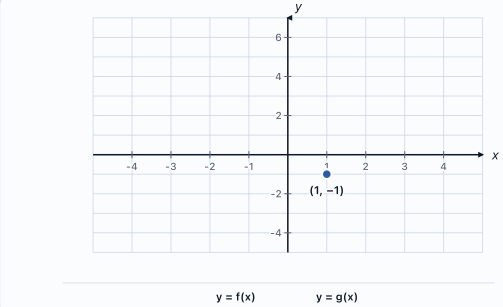
Mark Scheme & Examiner Solution (GAT Data Interpretation, Averages & Counting):

1. Use the sum formula for an arithmetic series: $S_n = \frac{n}{2}[2a_1 + (n-1)d]$. **[M1]**
2. Substitute $n = 15, a_1 = 7, d = 6$: $S_{15} = \frac{15}{2}[14 + 14 \times 6] = \frac{15}{2} \times 98 = 735$. **[M1][A1]**

- **Final Verified Answer:** 735.

5 MCQ

In GAT Data Interpretation, Averages & Counting, evaluate the exact definite integral $I_5 = \int_0^2 (18x^2 + 24x)\,\mathrm{d}x$.



A 120

B 120 + 1

C 49

D 96 ✓

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (GAT Data Interpretation, Averages & Counting):

1. Integrate term by term: $\int (18x^2 + 24x)\,\mathrm{d}x = [6x^3 + 12x^2]_0^2$. **[M1]**
2. Substitute the upper and lower limits: $(6(2)^3 + 12(2)^2) - 0 = 48 + 48 = 96$. **[M1][A1]**

- **Final Verified Answer:** 96.

6 MCQ

In GAT Data Interpretation, Averages & Counting, consider the quadratic function $f_6(x) = x^2 - 14x + 135$. Express $f_6(x)$ in the completed-square form $(x-h)^2 + k$ and compute the exact value of $k + f_6(14)$.

A 221 ✓

B 228

C 135

D 214

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (GAT Data Interpretation, Averages & Counting):

1. Complete the square: $x^2 - 14x + 135 = (x-7)^2 - 49 + 135 = (x-7)^2 + 86$, giving minimum value $k = 86$. **[M1]**
2. Evaluate at $x = 14$: $f_6(14) = (7)^2 + 86 = 135$. **[M1]**
3. Sum the two quantities: $k + f_6(14) = 86 + 135 = 221$. **[A1]**

- **Final Verified Answer:** 221.

7 MCQ

The graph illustrates the cubic model $y_7(x) = 2x^3 - 7x^2 + 33x - 18$ studied in GAT Data Interpretation, Averages & Counting. Find the instantaneous gradient $\frac{\mathrm{dy}}{\mathrm{dx}}$ of the tangent to the curve at $x = 8$.

A 417

B 305 ✓

C 272

D 67

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (GAT Data Interpretation, Averages & Counting):

1. Differentiate term by term using the power rule: $\frac{\mathrm{dy}}{\mathrm{dx}} = 6x^2 - 14x + 33$. **[M1]**
2. Substitute $x = 8$: $\left.\frac{\mathrm{dy}}{\mathrm{dx}}\right|_{x=8} = 6(64) - 14(8) + 33 = 305$. **[M1][A1]**

- **Final Verified Answer:** 305.

8 MCQ

In GAT Data Interpretation, Averages & Counting, two displacement vectors are given by $\mathbf{u}_8 = \langle 7, -7 \rangle$ and $\mathbf{v}_8 = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_8 \cdot \mathbf{v}_8 + \|\mathbf{u}_8\|^2$.

A 21

B 98

C 119 ✓

D 189

WORKED SOLUTION & VERIFICATION

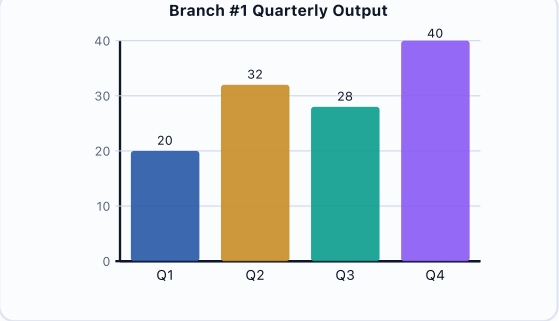
Mark Scheme & Examiner Solution (GAT Data Interpretation, Averages & Counting):

1. Compute the inner product: $\mathbf{u}_8 \cdot \mathbf{v}_8 = (7)(8) + (-7)(5) = 56 + (-35) = 21$. **[M1]**
2. Compute the squared norm: $\|\mathbf{u}_8\|^2 = 7^2 + (-7)^2 = 98$. **[M1]**
3. Combine: $21 + 98 = 119$. **[A1]**

- **Final Verified Answer:** 119.

9 MCQ

According to the bar chart shown for Branch #1, the quarterly outputs across Q1, Q2, Q3, and Q4 are 20, 32, 28, and 40 units, respectively. What is the arithmetic mean quarterly output?



A 28

B 30 ✓

C 60

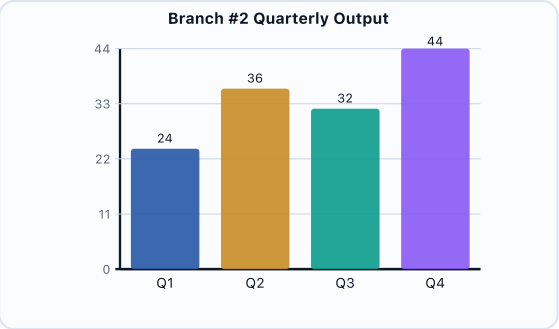
D 33

WORKED SOLUTION & VERIFICATION

Sum the four quarterly bars and divide by 4: $\frac{20+32+28+40}{4} = \frac{120}{4} = 30$.

10 MCQ

According to the bar chart shown for Branch #2, the quarterly outputs across Q1, Q2, Q3, and Q4 are 24, 36, 32, and 44 units, respectively. What is the arithmetic mean quarterly output?



A 32

B 34 ✓

C 68

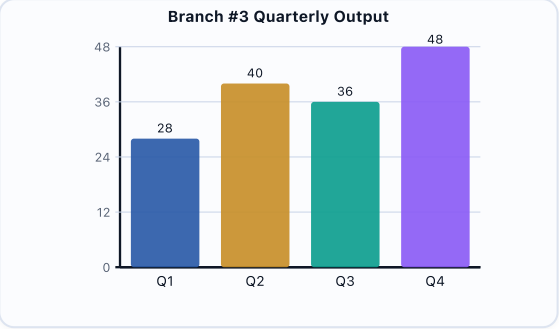
D 37

WORKED SOLUTION & VERIFICATION

Sum the four quarterly bars and divide by 4: $\frac{24+36+32+44}{4} = \frac{136}{4} = 34$.

11 MCQ

According to the bar chart shown for Branch #3, the quarterly outputs across Q1, Q2, Q3, and Q4 are 28, 40, 36, and 48 units, respectively. What is the arithmetic mean quarterly output?



A 76

B 41

C 36

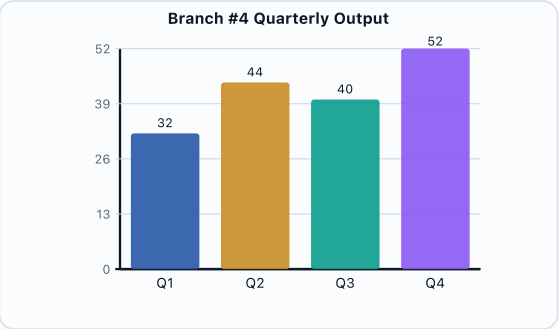
D 38 ✓

WORKED SOLUTION & VERIFICATION

Sum the four quarterly bars and divide by 4: $\frac{28+40+36+48}{4} = \frac{152}{4} = 38$.

12 MCQ

According to the bar chart shown for Branch #4, the quarterly outputs across Q1, Q2, Q3, and Q4 are 32, 44, 40, and 52 units, respectively. What is the arithmetic mean quarterly output?



A 84

B 40

C 42 ✓

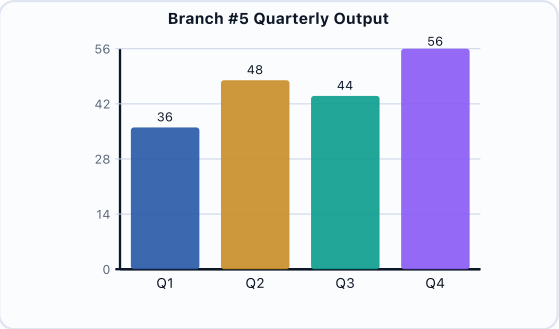
D 45

WORKED SOLUTION & VERIFICATION

Sum the four quarterly bars and divide by 4: $\frac{32+44+40+52}{4} = \frac{168}{4} = 42$.

13 MCQ

According to the bar chart shown for Branch #5, the quarterly outputs across Q1, Q2, Q3, and Q4 are 36, 48, 44, and 56 units, respectively. What is the arithmetic mean quarterly output?



A 44

B 46 ✓

C 92

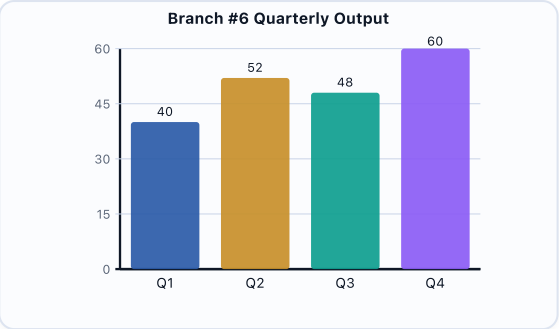
D 49

WORKED SOLUTION & VERIFICATION

Sum the four quarterly bars and divide by 4: $\frac{36+48+44+56}{4} = \frac{184}{4} = 46$.

14 MCQ

According to the bar chart shown for Branch #6, the quarterly outputs across Q1, Q2, Q3, and Q4 are 40, 52, 48, and 60 units, respectively. What is the arithmetic mean quarterly output?



A 53

B 50 ✓

C 48

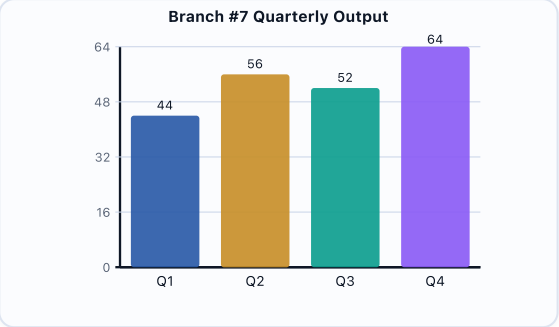
D 100

WORKED SOLUTION & VERIFICATION

Sum the four quarterly bars and divide by 4: $\frac{40+52+48+60}{4} = \frac{200}{4} = 50$.

15 MCQ

According to the bar chart shown for Branch #7, the quarterly outputs across Q1, Q2, Q3, and Q4 are 44, 56, 52, and 64 units, respectively. What is the arithmetic mean quarterly output?



A 57

B 54 ✓

C 52

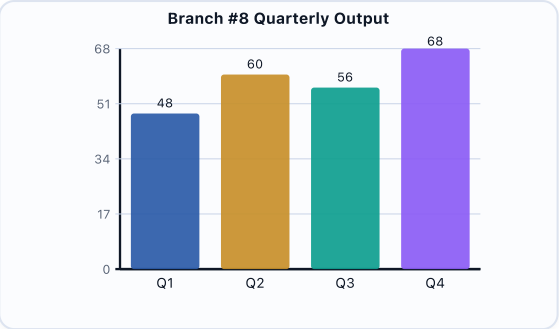
D 108

WORKED SOLUTION & VERIFICATION

Sum the four quarterly bars and divide by 4: $\frac{44+56+52+64}{4} = \frac{216}{4} = 54$.

16 MCQ

According to the bar chart shown for Branch #8, the quarterly outputs across Q1, Q2, Q3, and Q4 are 48, 60, 56, and 68 units, respectively. What is the arithmetic mean quarterly output?



A 116

B 58 ✓

C 56

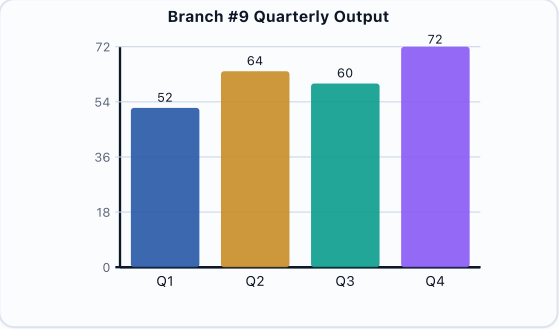
D 61

WORKED SOLUTION & VERIFICATION

Sum the four quarterly bars and divide by 4: $\frac{48+60+56+68}{4} = \frac{232}{4} = 58$.

17 MCQ

According to the bar chart shown for Branch #9, the quarterly outputs across Q1, Q2, Q3, and Q4 are 52, 64, 60, and 72 units, respectively. What is the arithmetic mean quarterly output?



A 62 ✓

B 65

C 60

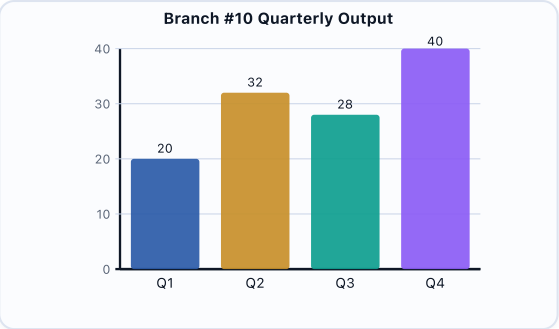
D 124

WORKED SOLUTION & VERIFICATION

Sum the four quarterly bars and divide by 4: $\frac{52+64+60+72}{4} = \frac{248}{4} = 62$.

18 MCQ

According to the bar chart shown for Branch #10, the quarterly outputs across Q1, Q2, Q3, and Q4 are 20, 32, 28, and 40 units, respectively. What is the arithmetic mean quarterly output?



A 30 ✓

B 60

C 33

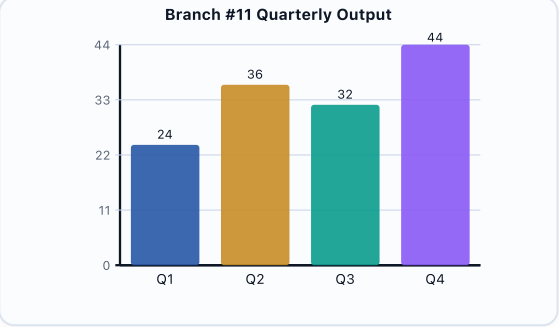
D 28

WORKED SOLUTION & VERIFICATION

Sum the four quarterly bars and divide by 4: $\frac{20+32+28+40}{4} = \frac{120}{4} = 30$.

19 MCQ

According to the bar chart shown for Branch #11, the quarterly outputs across Q1, Q2, Q3, and Q4 are 24, 36, 32, and 44 units, respectively. What is the arithmetic mean quarterly output?



A 68

B 32

C 37

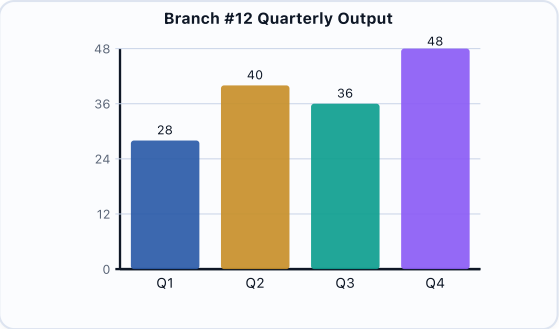
D 34 ✓

WORKED SOLUTION & VERIFICATION

Sum the four quarterly bars and divide by 4: $\frac{24+36+32+44}{4} = \frac{136}{4} = 34$.

20 MCQ

According to the bar chart shown for Branch #12, the quarterly outputs across Q1, Q2, Q3, and Q4 are 28, 40, 36, and 48 units, respectively. What is the arithmetic mean quarterly output?



A 76

B 41

C 36

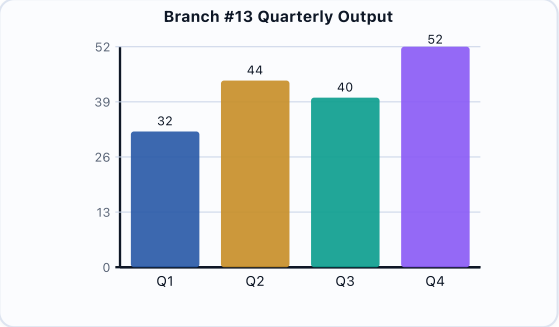
D 38 ✓

WORKED SOLUTION & VERIFICATION

Sum the four quarterly bars and divide by 4: $\frac{28+40+36+48}{4} = \frac{152}{4} = 38$.

21 MCQ

According to the bar chart shown for Branch #13, the quarterly outputs across Q1, Q2, Q3, and Q4 are 32, 44, 40, and 52 units, respectively. What is the arithmetic mean quarterly output?



A 40

B 84

C 42 ✓

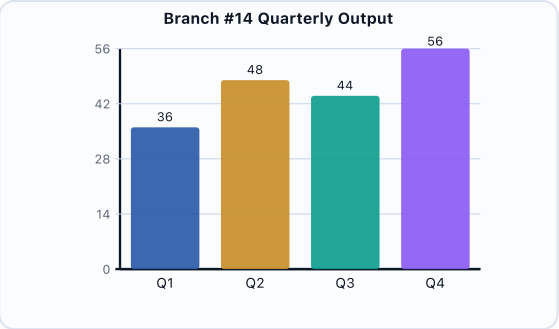
D 45

WORKED SOLUTION & VERIFICATION

Sum the four quarterly bars and divide by 4: $\frac{32+44+40+52}{4} = \frac{168}{4} = 42$.

22 MCQ

According to the bar chart shown for Branch #14, the quarterly outputs across Q1, Q2, Q3, and Q4 are 36, 48, 44, and 56 units, respectively. What is the arithmetic mean quarterly output?



A 49

B 92

C 44

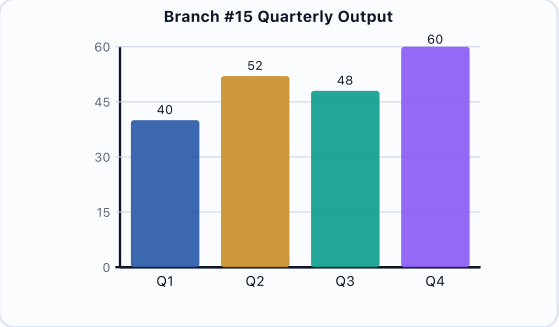
D 46 ✓

WORKED SOLUTION & VERIFICATION

Sum the four quarterly bars and divide by 4: $\frac{36+48+44+56}{4} = \frac{184}{4} = 46$.

23 MCQ

According to the bar chart shown for Branch #15, the quarterly outputs across Q1, Q2, Q3, and Q4 are 40, 52, 48, and 60 units, respectively. What is the arithmetic mean quarterly output?



A 48

B 53

C 50 ✓

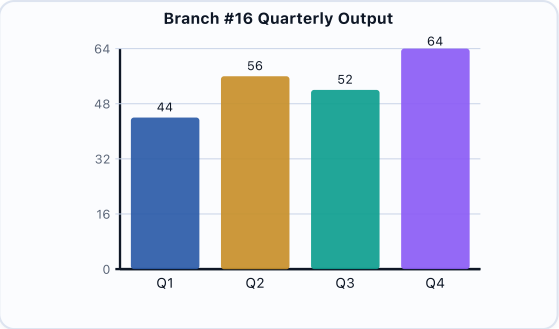
D 100

WORKED SOLUTION & VERIFICATION

Sum the four quarterly bars and divide by 4: $\frac{40+52+48+60}{4} = \frac{200}{4} = 50$.

24 MCQ

According to the bar chart shown for Branch #16, the quarterly outputs across Q1, Q2, Q3, and Q4 are 44, 56, 52, and 64 units, respectively. What is the arithmetic mean quarterly output?



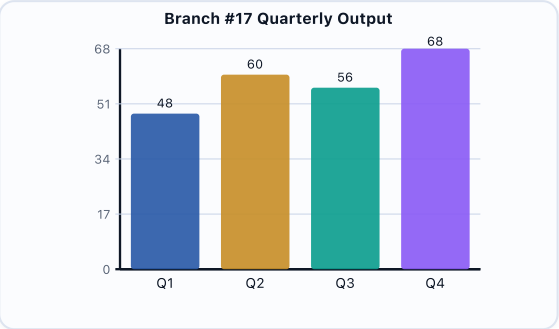
- **A** 54 ✓
- **B** 57
- **C** 52
- **D** 108

WORKED SOLUTION & VERIFICATION

Sum the four quarterly bars and divide by 4: $\frac{44+56+52+64}{4} = \frac{216}{4} = 54$.

25 MCQ

According to the bar chart shown for Branch #17, the quarterly outputs across Q1, Q2, Q3, and Q4 are 48, 60, 56, and 68 units, respectively. What is the arithmetic mean quarterly output?



- **A** 56
- **B** 61
- **C** 116
- **D** 58 ✓

WORKED SOLUTION & VERIFICATION

Sum the four quarterly bars and divide by 4: $\frac{48+60+56+68}{4} = \frac{232}{4} = 58$.